

Komla Gnona, PhD
(Local to Sunnyvale, CA and available for interviews immediately)

Summary:

- **Computational biologist** with hands-on experience analyzing large-scale **single-cell RNA-seq** and **perturbation-screen datasets** on **ex vivo** human cellular models.
- Analyzed large-scale multi-conditional single-cell RNA-seq (**scRNA-seq**) and perturbation-screen (**Perturb-seq**) datasets on iPSC-derived human neuronal ex vivo models to derive actionable insights for **drug discovery**.
- Built ML and **statistical models** classifying compound mechanisms of action from single-cell and phenotypic readouts; the work contributed to prioritization decisions on the program that is now in Phase 1 clinical trials.
- Built scalable **Nextflow** pipelines on **AWS** and **HPC (Slurm)** for terabyte-scale single-cell and multiomics datasets.
- Developed and optimized **bioinformatics** workflows for bulk, single-cell, and **spatial transcriptomics** on ex vivo neuronal cultures.
- Fluent in **Python** and **R** with a strong statistics foundation.
- Track record of collaborating tightly with wet-lab biologists and chemists to translate single-cell readouts into decisions that move drug-discovery programs forward.

Skills:

- **Core Expertise:** Single-Cell RNA-seq Analysis (Scanpy, Seurat), Perturbation-Screen Analysis (Perturb-seq style), Multi-Conditional scRNA-seq, Statistical Modeling on High-Dimensional Data, Multimodal Data Integration, Python Scientific Computing, Nextflow Pipeline Development, HPC and SLURM, Cross Functional Collaboration with Biologists and Chemists, Publication Track Record.
- **Single-Cell and Perturbation Analysis:** End-to-end scRNA-seq workflows using Scanpy (Python) and Seurat (R): QC, alignment, integration (Harmony, scVI concepts), clustering, cell-type annotation, differential expression, gene set enrichment, mechanism-of-action interpretation; hands-on experience with perturbation-based single-cell screens (Perturb-seq style); familiarity with CROP-seq and Sci-plex assay formats.
- **Statistics and Probabilistic Modeling:** Linear and generalized mixed-effects models, regularized regression, hypothesis testing on high-dimensional data, novel statistical-method development (peer reviewed in American Journal of Epidemiology); working familiarity with Bayesian and probabilistic modelling concepts.
- **Multimodal Data Integration:** Integrated single-cell, bulk transcriptomic, phenotypic, and clinical readouts at Sensorium and Amgen to inform target prioritization and biomarker discovery.
- **Programming:** Python (Pandas, NumPy, SciPy, scikit-learn, statsmodels, Scanpy, anndata, Biopython, Jupyter, PyTorch, TensorFlow), R (tidyverse, data.table, Bioconductor, Seurat, ggplot2, RMarkdown), SQL, Shell, Git.
- **Pipelines and Infrastructure:** Nextflow for reproducible bioinformatics pipelines, Slurm and Sun Grid Engine (SGE) on Linux HPC clusters, AWS (S3, EC2, batch compute), Google Cloud Platform (GCP); version control, code review, and reproducible documentation.

Education & Publications:

- PhD, Computational Biology, The Ohio State University, OH, Mar 2022
- BS, Mathematics, The Ohio State University, OH, Aug 2015
- Gnona KM, et al. Revisiting the Wald Test in Small Case-Control Studies. American Journal of Epidemiology, 2022.
- Stewart WCL, Gnona KM, et al. Genetic Prediction of Neonatal Complications Using Exome-Wide Variation. Pediatric Research, 2020.

Work experience:

Amgen, South San Francisco, CA

Apr 2025 – May 2026

Senior Scientist, Translational Analytics

- Analyzed **multi-omics** and **clinical datasets** to identify mechanistic drivers of immune-mediated disease, integrating heterogeneous data types for translational decision-making.
- Built and benchmarked predictive **ML** models on high-dimensional biological data with rigorous validation and error-mode analysis.
- Communicated technical findings to clinical, translational, and discovery stakeholders.

Sensorium Therapeutics, Boston, MA

Sep 2022 – Apr 2025

R&D Scientist II, Computational Biology (Sep 2024 - Apr 2025)

- Analyzed large-scale multi-conditional **single-cell RNA-seq** and **perturbation-screen datasets** on iPSC-derived human neuronal **ex vivo** models to derive actionable insights for drug discovery.
- Built **ML** and **statistical models** classifying compound mechanisms of action from **single-cell** and **phenotypic readouts**; the work contributed to prioritization decisions on the program that is now in Phase 1 clinical trials.
- Built scalable **Nextflow** pipelines on AWS and **HPC (Slurm)** for terabyte-scale single-cell and multiomics datasets; enforced reproducibility through version control and structured code review.
- Collaborated closely with biologists, chemists, and electrophysiologists on experimental design and interpretation.
- Co-wrote and helped secure SBIR Phase I research funding.

R&D Scientist I, Computational Biology (Sep 2022 – Sep 2024)

- Developed and optimized **bioinformatics** workflows for bulk, single-cell, and spatial **transcriptomics** on ex vivo neuronal cultures.
- Conducted integrative **multi-omics** analyses across gene expression, phenotype, and functional readouts.
- Improved data reliability through normalization, batch-effect correction, and feature engineering.

Henry M. Jackson Foundation, Bethesda, MD

Mar 2022 – Aug 2022

Research Scientist

- Built **NGS** pipelines for viral and host genomic sequence data; applied **statistical** and survival models to identify mutation-driven patterns.
- Conducted sequence homology (BLAST, HMMER) and phylogenomic analyses for mutation-impact assessment.

The Ohio State University, Columbus, OH

Aug 2016 – Mar 2022

Graduate Research Associate, PhD in Computational Biology (Statistical Genetics)

- Built **statistical and computational frameworks** for GWAS, variant calling, and predictive risk stratification on **large-scale sequencing** cohorts.
- Developed novel **statistical-testing methodology** for case-control studies, peer-reviewed in the American Journal of Epidemiology (2022).
- Managed **HPC** infrastructure (**Slurm**, **SGE**) for terabyte-scale parallel computation.